AutoMetric AI

AI Assisted Car Valuation
and Maintenance

Built for Pakistan's used car market

Price estimates in PKR

Android and web

ABSTRACT

Buying or selling a used car often begins with one question: what is a reasonable asking price? AutoMetric AI uses vehicle listings from Pakistan to estimate prices from a car's specifications and mileage. Available on Android and web, it gives buyers and sellers a useful starting point for comparison and negotiation. Owners can also keep service records and plan upcoming maintenance in the same app.

PROBLEM STATEMENT

Similar-looking cars can carry very different asking prices. Checking listings manually takes time, while differences in age, mileage and variant make comparisons difficult. Service records and maintenance deadlines are also easy to lose track of.

AIM AND OBJECTIVES

Aim: Provide a locally relevant price reference and a simple way to manage vehicle upkeep.

- Estimate used car prices using vehicle details and local listing data.
- Let users save valuations for later comparison.
- Organize service records and track upcoming maintenance.

PROPOSED SOLUTION

Enter a car's details to view an estimated price in PKR. Sign in to save valuations or manage a separate garage with multiple cars. Record services and odometer readings to review upcoming maintenance.

METHODOLOGY

01 **Collect:** Gather used car listings from PakWheels.

02 **Prepare:** Clean records and standardize vehicle details.

03 **Develop:** Compare prediction models and select XGBoost.

04 **Connect:** Serve predictions through FastAPI and package the backend with Docker.

05 **Build:** Bring valuation and maintenance features together in Flutter, with Firebase for user records.

RESULTS

9,849

prepared vehicle records

- Price estimates on Android and web.
- Saved valuations and a garage for multiple cars.
- Service history and reminders using dates and updated mileage.

Estimates reflect advertised prices and do not replace a vehicle inspection.

TECHNOLOGIES USED

Flutter and Dart
Android and web interface

Python and XGBoost
Data preparation and price prediction

Firebase
Accounts and personal records

FastAPI and Docker
Prediction service and packaging

SUMMARY

AutoMetric AI gives buyers and sellers a data-based starting point for price discussions, while helping owners keep vehicle care organized.

PROJECT DEMO

Demo QR

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